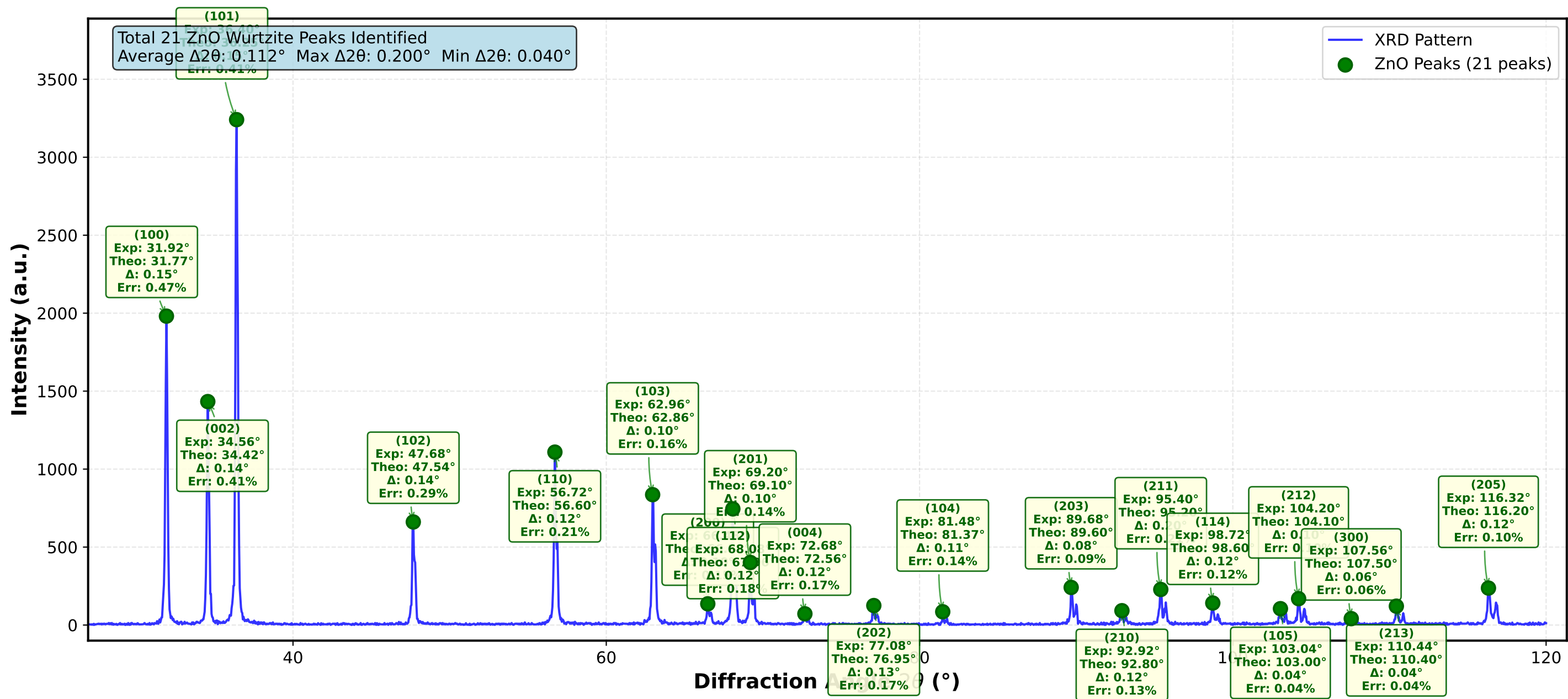
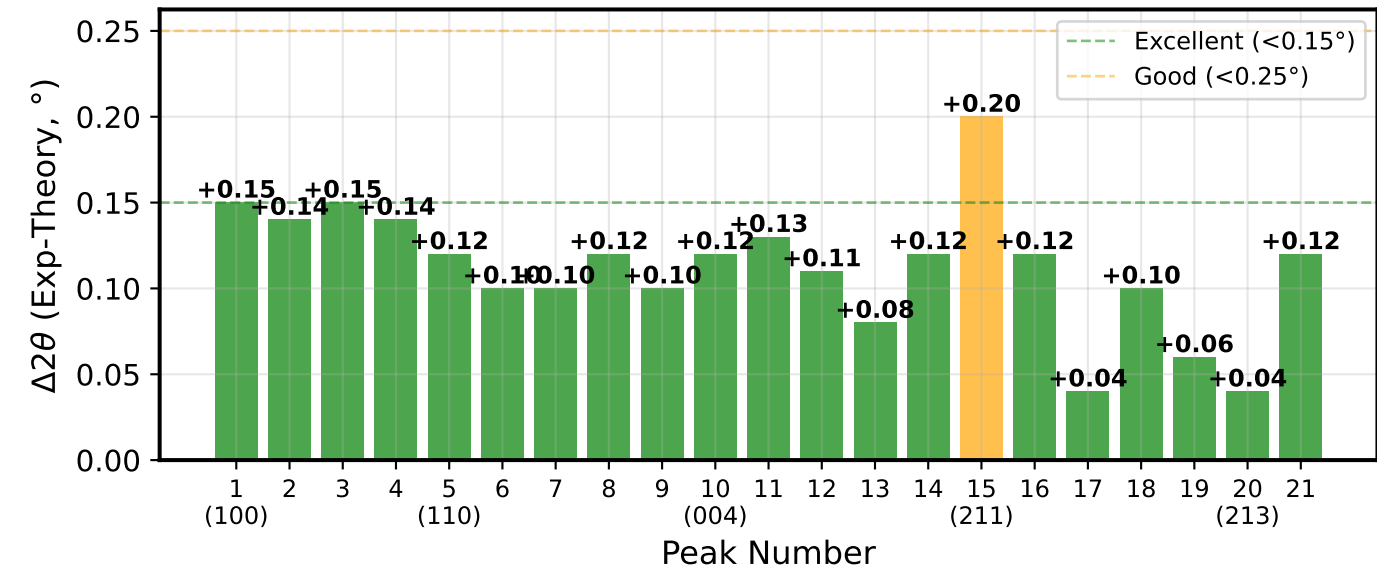


Complete XRD Analysis of ZnO Wurtzite Structure - All 21 Peaks Identified

XRD Pattern of ZnO Wurtzite Structure with All 21 Peaks Identified



Peak Position Deviation for All 21 Peaks



Detailed Peak Matching Analysis						
No.	2 θ _exp	2 θ _theo	$\Delta 2\theta$	hkl	Intensity	Error(%)
1	31.92	31.77	0.15	100	1981	0.472
2	34.56	34.42	0.14	002	1433	0.407
3	36.40	36.25	0.15	101	3241	0.414
4	47.68	47.54	0.14	102	661	0.294
5	56.72	56.60	0.12	110	1109	0.212
6	62.96	62.86	0.10	103	836	0.159
7	66.48	66.38	0.10	200	136	0.151
8	68.08	67.96	0.12	112	745	0.177
9	69.20	69.10	0.10	201	401	0.145
10	72.68	72.56	0.12	004	72	0.165
11	77.08	76.95	0.13	202	125	0.169
12	81.48	81.37	0.11	104	85	0.135
13	89.68	89.60	0.08	203	241	0.089
14	92.92	92.80	0.12	210	92	0.129
15	95.40	95.20	0.20	211	228	0.210
16	98.72	98.60	0.12	114	141	0.122
17	103.04	103.00	0.04	105	105	0.039
18	104.20	104.10	0.10	212	169	0.096
19	107.56	107.50	0.06	300	41	0.056
20	110.44	110.40	0.04	213	121	0.036
21	116.32	116.20	0.12	205	237	0.103

Statistics:

- Average $\Delta 2\theta$: 0.112°
- Max $\Delta 2\theta$: 0.200° (Peak 15)
- Min $\Delta 2\theta$: 0.040° (Peak 17)

Lattice Parameters:

- a = $2.8193 \pm 0.0228 \text{ \AA}$
- c = $5.1931 \pm 0.0066 \text{ \AA}$
- c/a ratio = 1.8419